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1**Functions and functional zones**

COMP

You should be able to: State the functions of the respiratory system and distinguish the conducting zone from the respiratory zone by structure and by role in gas exchange.

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2**Pressure gradients and airflow**

COMP

You should be able to: Apply Boyle's law to inspiration and expiration and relate alveolar pressure to atmospheric pressure at each point in a quiet breath.

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3**Intrapleural pressure**

COMP

You should be able to: Explain why intrapleural pressure is subatmospheric and predict what happens to the lung and chest wall when the pleural seal is broken.

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4**Compliance and elastic recoil**

COMP

You should be able to: Define lung compliance and elastic recoil and predict the effect of fibrosis and of emphysema on each.

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5**Surfactant and surface tension**

COMP

You should be able to: Explain how surfactant lowers surface tension and why its absence causes alveolar collapse in the premature newborn.

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6**Airway resistance**

COMP

You should be able to: Identify the main determinant of airway resistance and predict the effect of bronchoconstriction, mucus, and sympathetic stimulation on airflow.

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7**Lung volumes and capacities**

COMP

You should be able to: Define tidal volume, inspiratory and expiratory reserve volume, residual volume, vital capacity, and total lung capacity and read each from a spirogram.

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8**Obstructive and restrictive patterns**

COMP

You should be able to: Use the ratio of forced expiratory volume in one second to forced vital capacity to classify a spirometry result as obstructive or restrictive.

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9**Alveolar ventilation and dead space**

COMP

You should be able to: Calculate minute ventilation and alveolar ventilation and explain why slow deep breathing ventilates the alveoli better than rapid shallow breathing.

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10**Spirometry**

COMP

You should be able to: Record a spirogram, calculate the lung volumes and capacities available from it, and compare the results with predicted values.

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11**Partial pressures**

COMP

You should be able to: Apply Dalton's law to calculate the partial pressure of oxygen in inspired and alveolar air and state the normal partial pressures in alveolar air, arterial blood, and venous blood.

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12**Diffusion at the respiratory membrane**

COMP

You should be able to: Apply the determinants of diffusion to the respiratory membrane and predict the effect of edema, fibrosis, and emphysema on gas transfer.

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13**Ventilation perfusion matching**

COMP

You should be able to: Explain how local hypoxic vasoconstriction and bronchiolar responses match ventilation to perfusion and predict the consequence of a mismatch.

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14**Oxygen transport and the oxyhemoglobin curve**

COMP

You should be able to: Explain the sigmoid shape of the oxyhemoglobin dissociation curve and state the physiological advantage of the steep and the flat portions.

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15**Curve shifts**

COMP

You should be able to: Predict the direction of the oxyhemoglobin curve shift produced by a change in pH, carbon dioxide, temperature, or 2,3 bisphosphoglycerate and state what the shift means for tissue oxygen delivery.

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16**Carbon dioxide transport**

COMP

You should be able to: Name the three forms in which carbon dioxide is carried, state the approximate percentage of each, and trace the chloride shift in a systemic capillary.

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17**Bohr and Haldane effects**

COMP

You should be able to: Distinguish the Bohr effect from the Haldane effect and explain how the two work together at the tissue and at the lung.

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18**Oxygen content versus partial pressure**

COMP

You should be able to: Distinguish oxygen content from oxygen partial pressure and saturation and explain why anemia and carbon monoxide poisoning reduce delivery differently.

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19**Respiratory centers**

COMP

You should be able to: Identify the medullary and pontine respiratory centers and explain how the basic breathing rhythm is generated and modified.

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20**Chemoreceptor control**

COMP

You should be able to: Compare central and peripheral chemoreceptors by stimulus and explain why carbon dioxide is the primary minute to minute drive to breathe.

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21**Ventilation in exercise and at altitude**

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You should be able to: Predict the ventilatory response to exercise and to acute and chronic altitude exposure and explain the acclimatization changes.

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22**Ventilatory response testing**

COMP

You should be able to: Measure the change in breathing rate and depth after hyperventilation, breath holding, and rebreathing and explain each result by the change in carbon dioxide.

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23**Buffer systems**

COMP

You should be able to: Compare the bicarbonate, phosphate, and protein buffer systems by location and speed and explain why bicarbonate is the dominant extracellular buffer.

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24**Respiratory control of pH**

COMP

You should be able to: Explain how a change in ventilation shifts the carbon dioxide and bicarbonate equilibrium and therefore plasma pH.

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